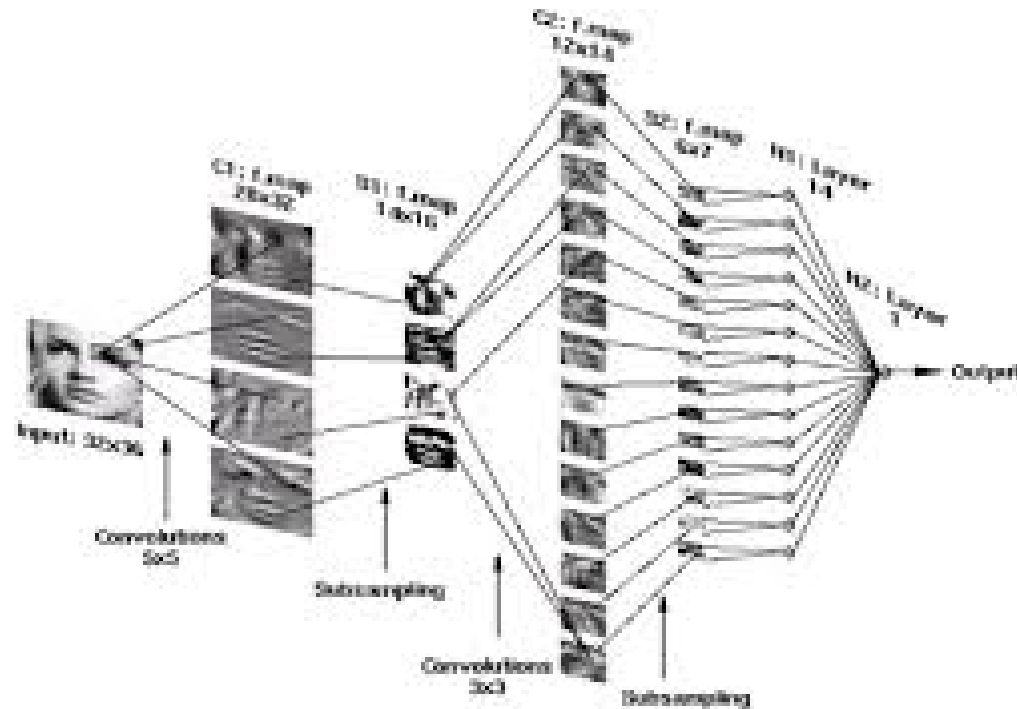


Mathematics of Machine Learning

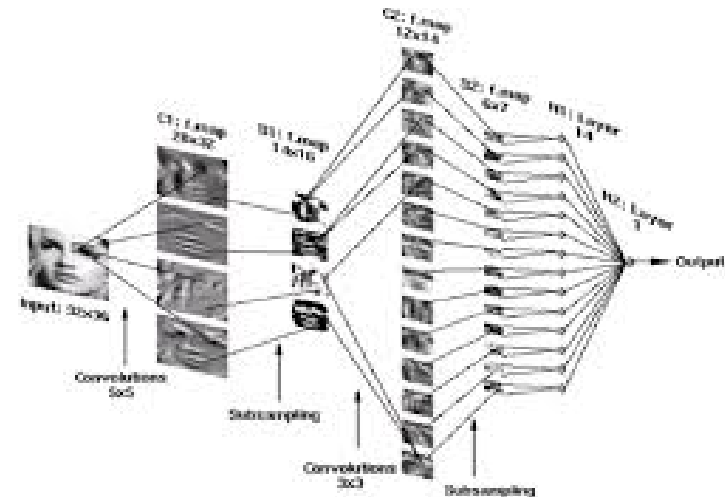
Introduction



Course Details and Topics

Outline

1. **Why are you doing this course?**
2. What we assume that you know



Orientation

- This course addresses three cohorts of students:
 - ★ Third year Artificial Intelligence Students
 - ★ MSc in Machine Learning Students
 - ★ Volunteers
- It is dual coded (AICE3001/AICE6002), with subtle difference in expectations reflected in slightly different evaluation
- Notes are on Moodle and

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- LLMs can help us a lot, but if you don't trust yourself you are disempowered
- This course is aimed at building up trust in your understanding of how machine learning works

Why Mathematics?

- Learning machines are notoriously difficult to understand
- Mathematics allows us to reason rigorously
- Posing question mathematically and seeking mathematical solutions provide allow us to build a strong intuition
- Mathematics also allow us to describe the world precisely
- It also provides an arsenal of tools that turn out to be hugely useful

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Cracking the Code

- Mathematics is a language that takes time to master
- For some it is easy, for others it will be a struggle
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Starting from Different Places

- You all start from different places
- Some of you may have a degree in mathematics
- For others it may have been your least favourite subject
- I invite strong mathematicians to become even stronger
- For students starting from a weak base, judge yourselves by your progress, not by comparing yourself to others

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The Journey

- We are going to visit lots of aspects of mathematics that have proved fruitful in machine learning
- Some of this will be very concrete, while in other cases by look at a higher level of abstraction we can generalise methods across different application areas
- We will cover, calculus, vector spaces, convexity, optimisation, entropy, probability, statistics and much else besides
- The journey will pay you off, these are the foundations of most of machine learning and will give you super-powers that are hard to acquire without mathematics

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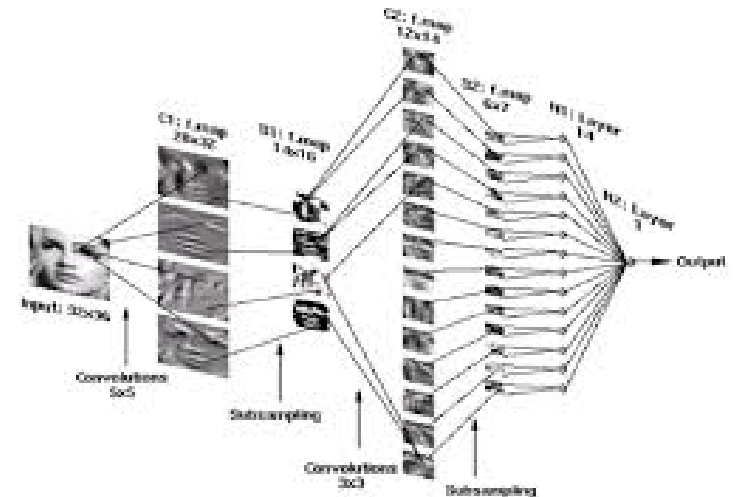
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What we Assume you Know

- You should all have A-level mathematics and have done some mathematics at Uni
- You should know what a function is, e.g. $f(x) = x^2 + 3$
- You should be able to do simple algebraic manipulations
- And some elementary calculus

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Vectors

- Ordinary vectors (what mathematicians call elements of $\mathbb{R}^n$) are much used in machine learning and should be familiar
- We will use bold symbols to represent **column vectors**, e.g.

$$\mathbf{x} = \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix}$$

- We denote **row vectors** as the **transpose** of column vectors
 $\mathbf{x}^T = (1.2, 2.3, 4.7)$
- We care whether vectors are row or column vectors as it allows us to write equations involving vectors succinctly and unambiguously

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Matrices

- Matrices are blocks of number that define linear mappings between vectors, e.g. $\mathbf{M} \in \mathbb{R}^{2 \times 3}$

$$\mathbf{M} = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix}$$

- An $m \times n$ matrix (i.e. $\mathbf{M} \in \mathbb{R}^{m \times n}$) has m rows and n columns
- An $m \times n$ matrix acting on a vector in $\mathbb{R}^n$ transforms it to $\mathbb{R}^m$ vector, e.g.

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Composition of Mappings

- Matrices, $\mathbf{A}$, $\mathbf{B}$, etc. represent mappings between spaces
- Provided the matrices are the correct size we can compose these mappings, e.g. if $\mathbf{A} \in \mathbb{R}^{\ell \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$ then $\mathbf{AB}$ represents the effect of
 - ★ first applying $\mathbf{B}$ to vectors in $\mathbb{R}^n$ to obtain vectors in $\mathbb{R}^m$
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Matrix Multiplication

- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)

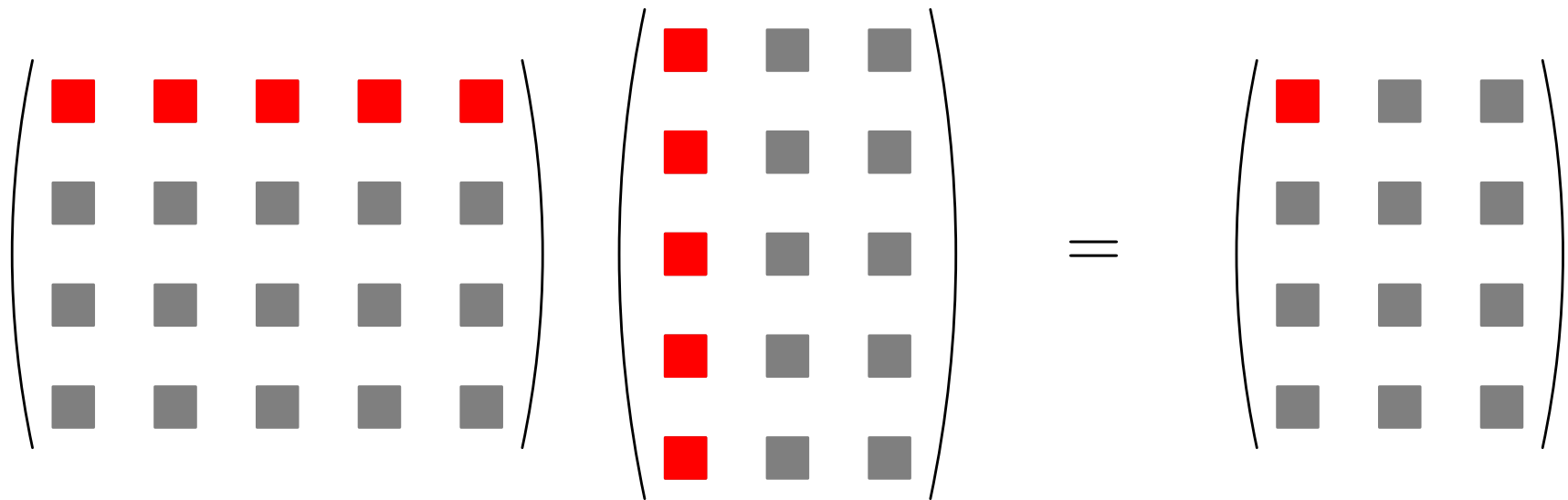
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The diagram illustrates matrix multiplication using a visual representation of matrices as grids of squares. The first matrix, representing $\mathbf{A}$, is a 5x5 grid with 5 rows and 5 columns. The second matrix, representing $\mathbf{B}$, is a 5x3 grid with 5 rows and 3 columns. An equals sign follows, leading to the resulting matrix $\mathbf{C}$, which is a 5x3 grid with 5 rows and 3 columns. This visualizes the operation $\mathbf{AB} = \mathbf{C}$.

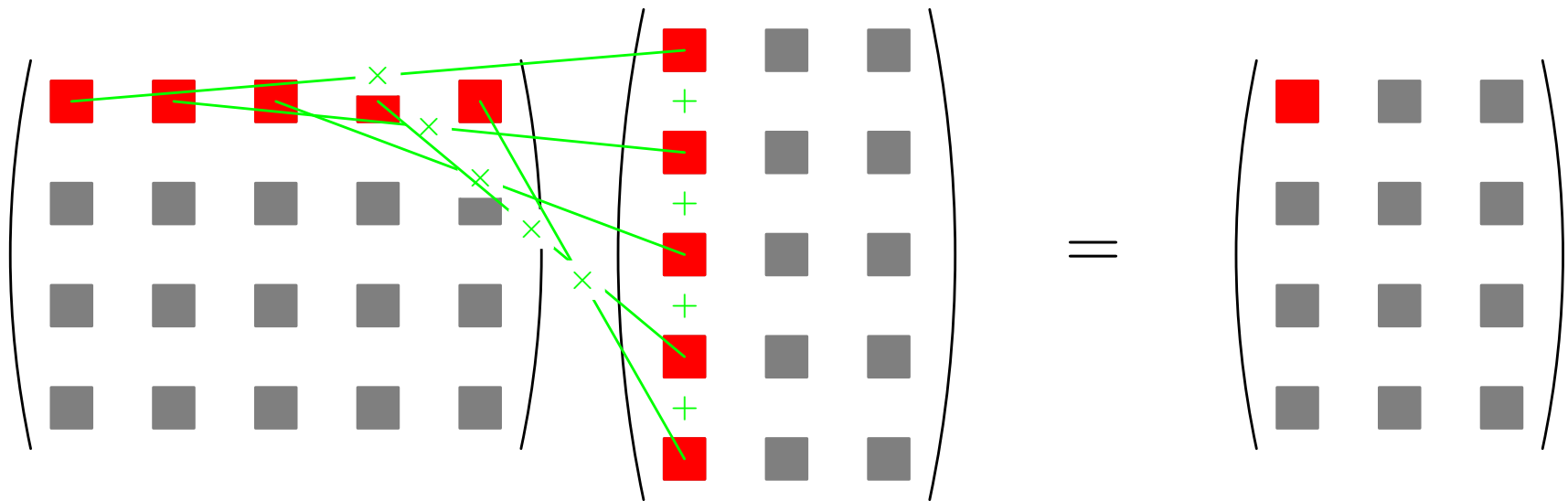
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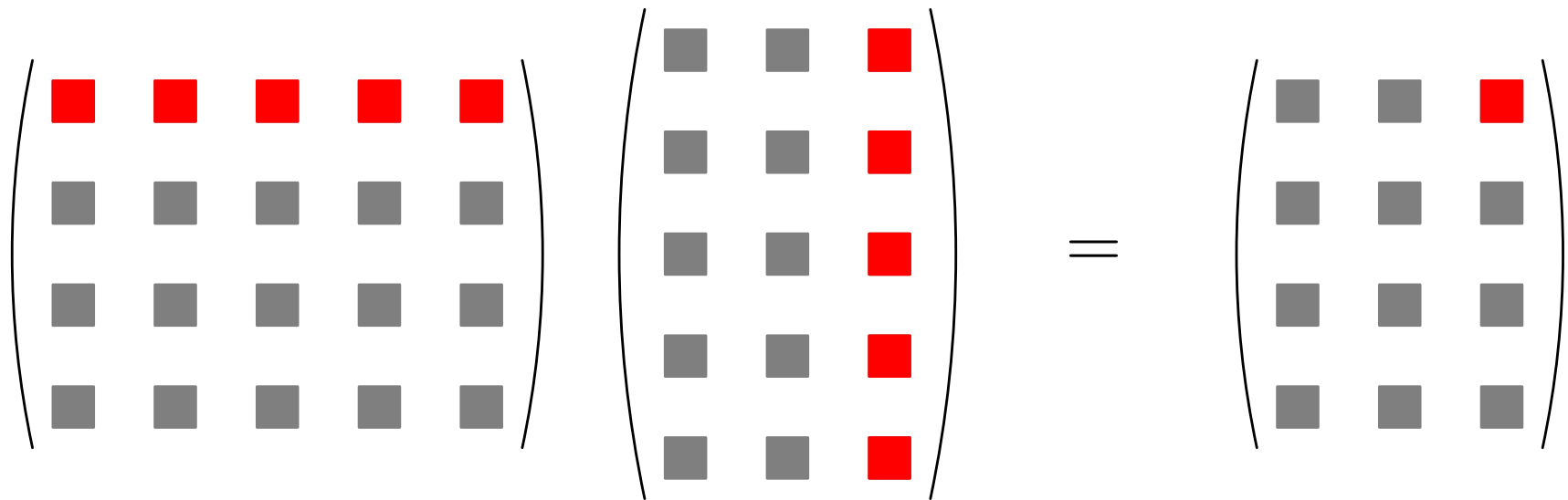
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Gray	Gray	Gray	Gray	Gray	Gray	Gray	Gray	Gray
Red	Red	Red	Red	Red	Gray	Gray	Gray	Gray
Gray	Gray	Gray	Gray	Gray	Gray	Gray	Gray	Gray
Gray	Gray	Gray	Gray	Gray	Gray	Gray	Gray	Gray

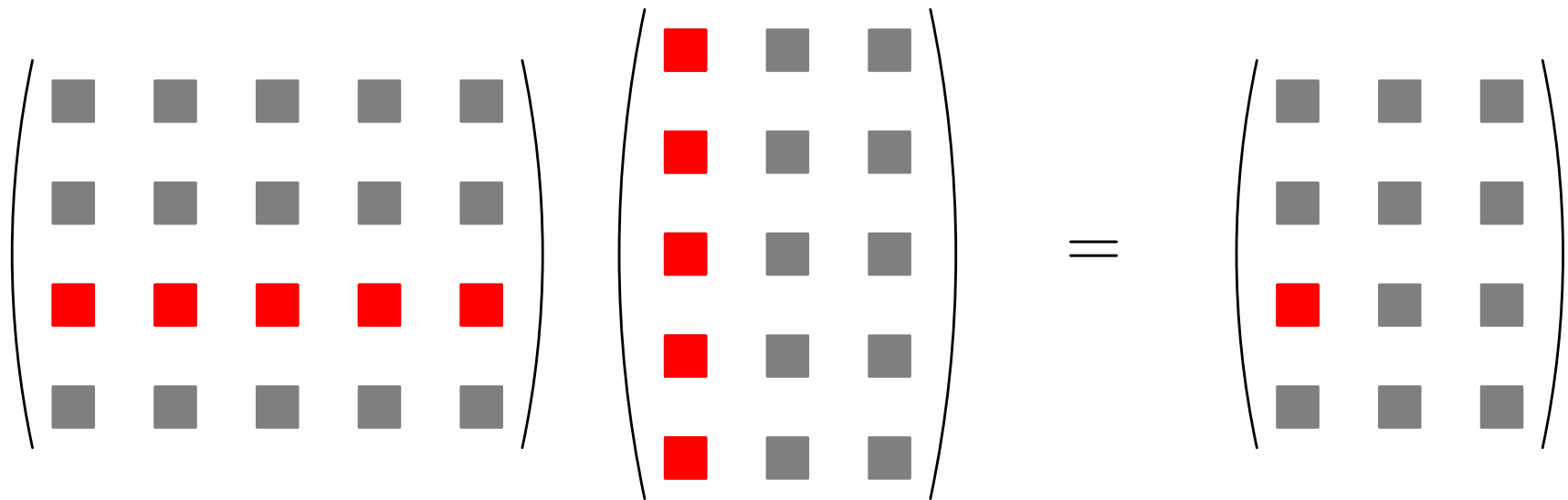
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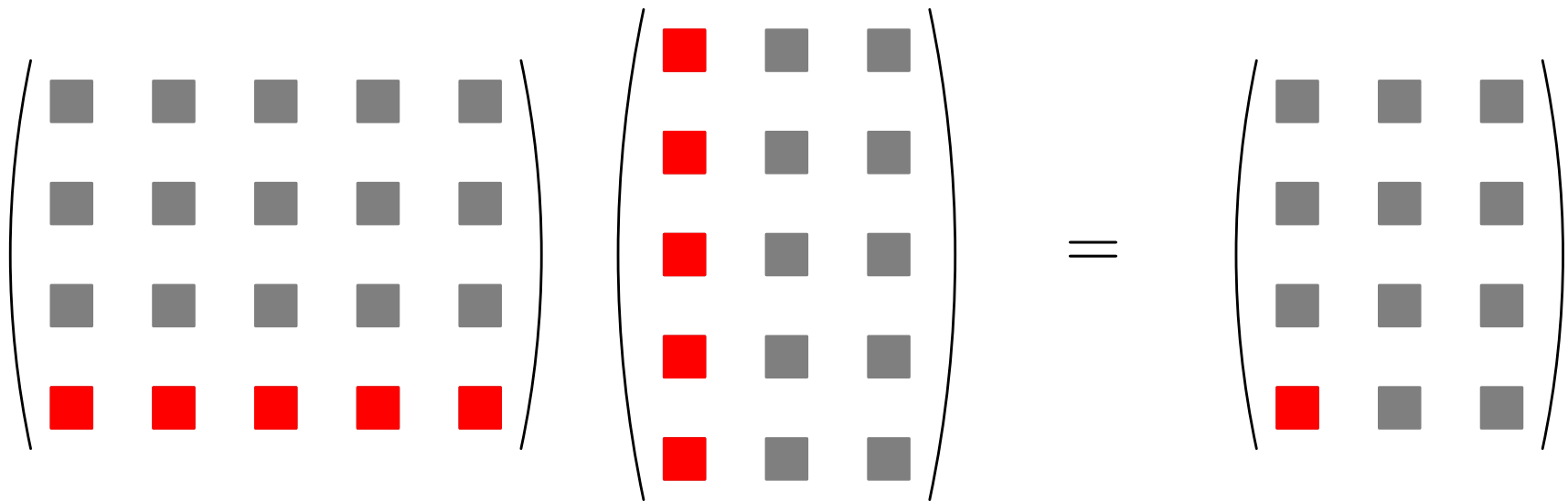
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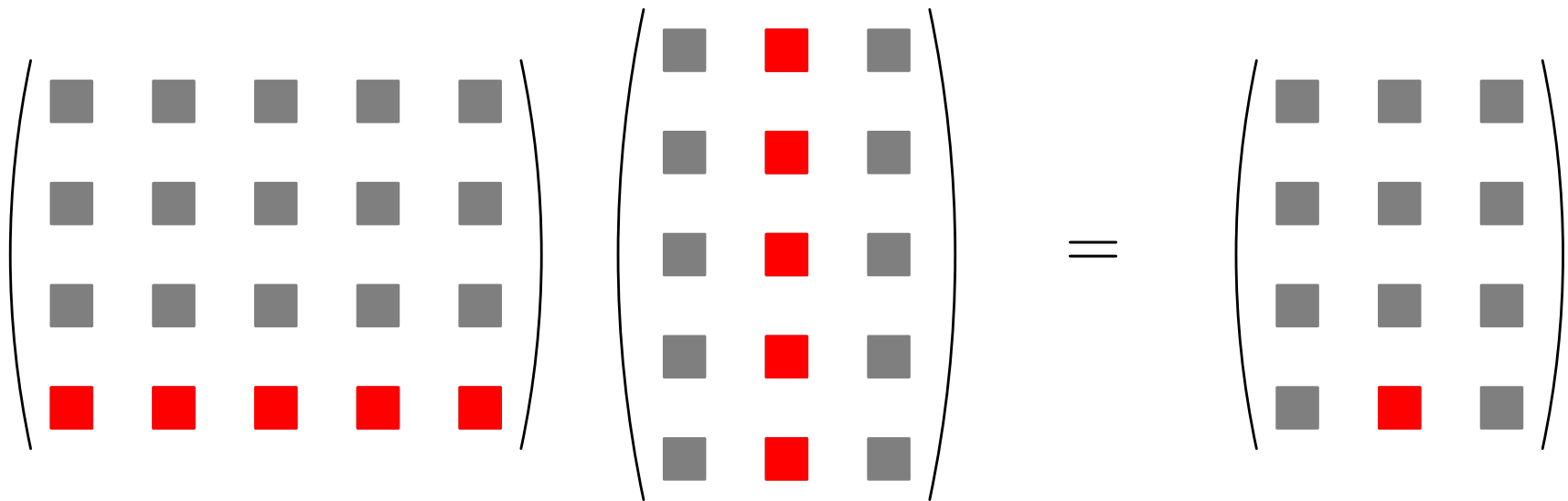
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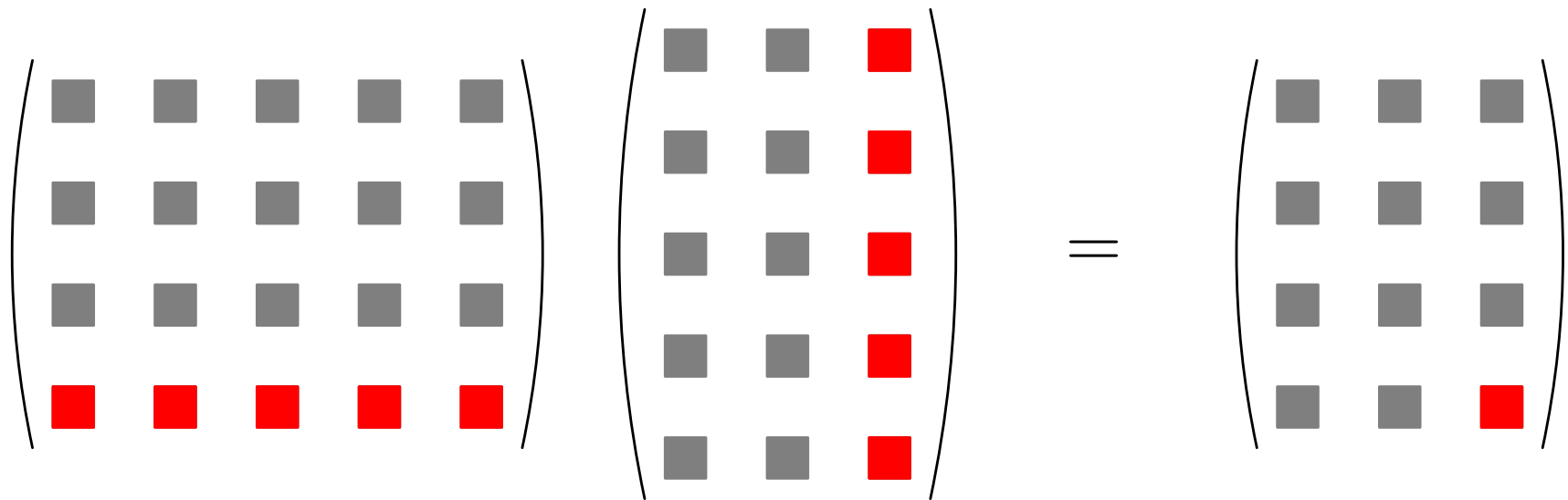
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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is a 4x5 grid with 4 rows and 5 columns. The first three rows are gray, and the fourth row is red. The second matrix is a 5x3 grid with 5 rows and 3 columns. The first four rows are gray, and the fifth row is red. The result is a 4x3 grid with 4 rows and 3 columns. The first three rows are gray, and the fourth row is red. The matrices are enclosed in large parentheses, and an equals sign is placed between the second and third matrices.

Matrix Multiplication

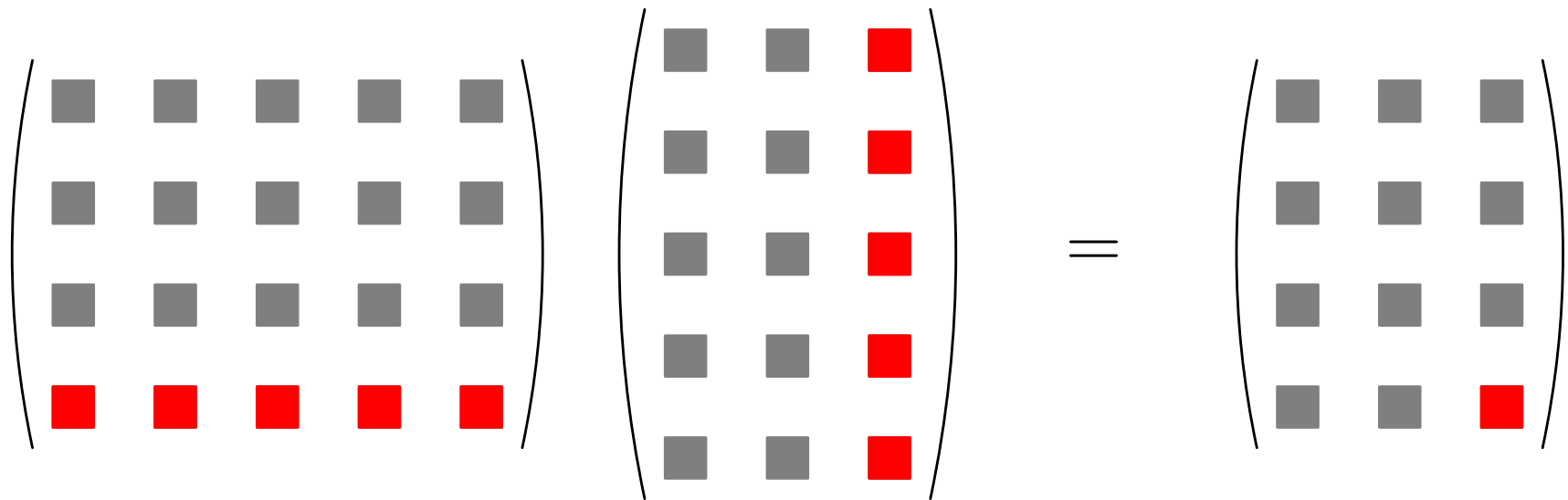
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- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
- But in general not commutative $\mathbf{AB} \neq \mathbf{BA}$

Transpose

- We often encounter transposes of a matrix $\mathbf{M}^T$ where we have exchanged the rows and columns: $[\mathbf{M}^T]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^T)^T = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^T$
- Since 1×1 matrices are always symmetric

$$\mathbf{a}^T \mathbf{b} = (\mathbf{a}^T \mathbf{b})^T = \mathbf{b}^T \mathbf{a}$$

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- We often encounter transposes of a matrix $\mathbf{M}^\top$ where we have exchanged the rows and columns: $[\mathbf{M}^\top]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^\top)^\top = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$

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$$\mathbf{a}^\top \mathbf{b} = (\mathbf{a}^\top \mathbf{b})^\top = \mathbf{b}^\top \mathbf{a}$$

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- Clear $(\mathbf{M}^\top)^\top = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^\top$
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$$\begin{aligned} \mathbf{a}^\top \mathbf{b} &= (\mathbf{a}^\top \mathbf{b})^\top = \mathbf{b}^\top \mathbf{a} \\ \mathbf{a}^\top \mathbf{M} \mathbf{b} &= \mathbf{b}^\top \mathbf{M}^\top \mathbf{a} \end{aligned}$$

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 4x5 matrix is represented by a grid of 20 gray squares enclosed in large parentheses. To its right is a 4x1 column vector, represented by a single column of 4 gray squares enclosed in large parentheses. An equals sign follows, and to the right is the resulting 4x1 column vector, also represented by a single column of 4 gray squares enclosed in large parentheses.

Matrix-Vector Multiplication

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The diagram illustrates the concept of treating a column vector as a matrix for multiplication. It shows a 4x5 matrix (with 5 red squares in the top row and 4 gray squares in each of the other three rows) multiplied by a 5x1 column vector (with 5 red squares). This is equated to a single 4x1 column vector (with 1 red square in the top row and 3 gray squares below it), representing the result of the multiplication.

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 5x5 matrix is shown with gray squares in all positions except the second row, which contains five red squares. This matrix is multiplied by a 5x1 column vector, also shown on the left, which consists of five red squares. An equals sign follows, and on the right is a 5x1 column vector with five elements: the first, third, fourth, and fifth elements are gray squares, while the second element is a red square. This represents the result of the multiplication, where only the second row of the original matrix is active because the second element of the vector is non-zero.

Matrix-Vector Multiplication

- We can treat a column vector as a $n \times 1$ matrix
- Then matrix-vector multiplication can be treated the same as matrix multiplication

The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 5x5 matrix is shown with 4 gray squares and 1 red square in each row. This matrix is multiplied by a 5x1 column vector consisting of 5 red squares. The result is a 5x1 column vector consisting of 4 gray squares and 1 red square. The entire equation is enclosed in large parentheses.

$$\begin{pmatrix} \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} \\ \text{gray} \\ \text{gray} \\ \text{red} \\ \text{gray} \end{pmatrix}$$

Matrix-Vector Multiplication

- We can treat a column vector as a $n \times 1$ matrix
- Then matrix-vector multiplication can be treated the same as matrix multiplication

The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 5x5 matrix (represented by a grid of 25 squares, with the bottom row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of 5 red squares). The result is a 5x1 column vector (represented by a single column of 5 squares, with the bottom square highlighted in red). The entire equation is enclosed in large parentheses.

$$\begin{pmatrix} \text{5x5 grid of squares} \end{pmatrix} \begin{pmatrix} \text{5x1 column of red squares} \end{pmatrix} = \begin{pmatrix} \text{5x1 column of squares} \end{pmatrix}$$

Inner Products

- The dot or inner-product between two vectors with the same number of elements, can be written $a^T b$

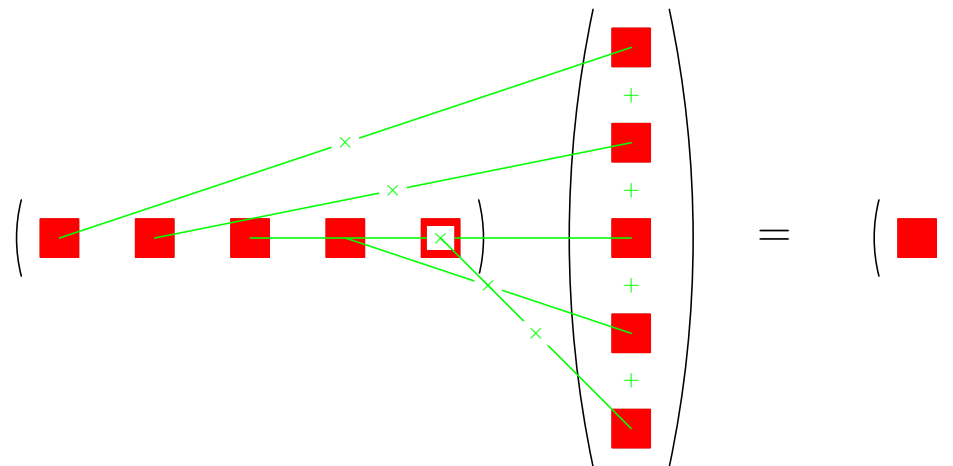
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$$\left(\begin{array}{ccccc} \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare \end{array} \right) = \left(\begin{array}{c} \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \end{array} \right) = \left(\blacksquare \right)$$

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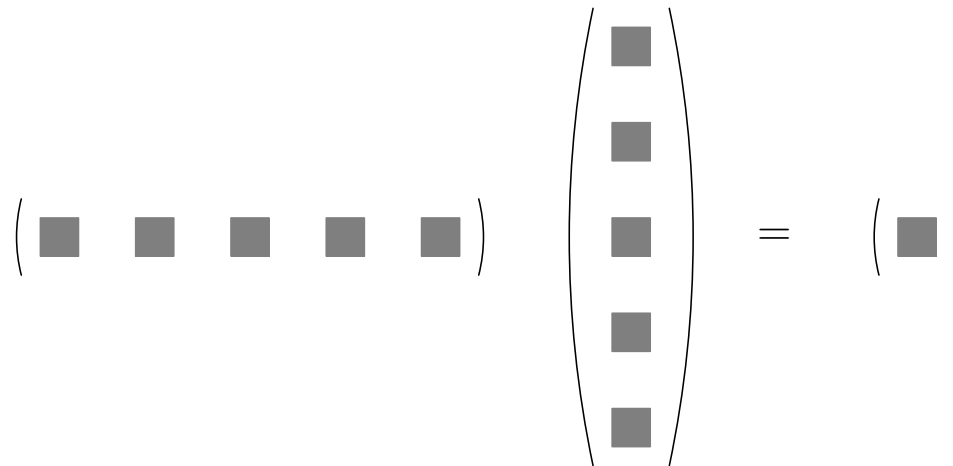
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- Inner products encode angles between vectors

$$\mathbf{a}^T \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$$

Outer Products

- We can also define the outer-product between any size vectors, ab^T

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- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is represented by a large left parenthesis followed by four gray squares stacked vertically, and a large right parenthesis. This is multiplied by a row vector b^T of size 1x3, represented by a large left parenthesis followed by three gray squares, and a large right parenthesis. An equals sign follows, leading to the resulting 4x3 matrix. This matrix is shown as a large left parenthesis followed by four rows, each containing three gray squares, and a large right parenthesis.

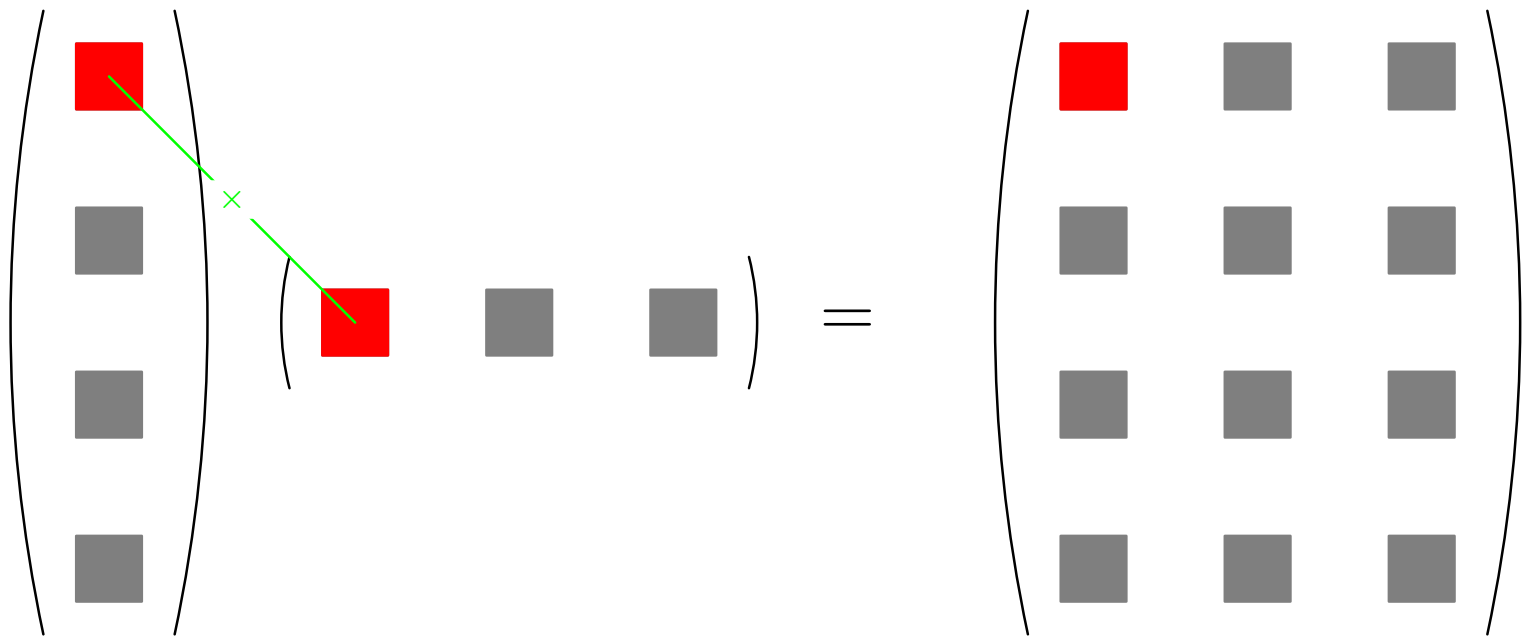
Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is shown, with its top element highlighted in red. This is multiplied by a row vector b^T of size 1x3, also with its first element highlighted in red. The result is a 4x3 matrix where each row is a scalar multiple of the vector b^T . The first row of the resulting matrix has its first element highlighted in red, while the other rows have gray elements.

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4x1 is shown, with its top element highlighted in red. This is multiplied by a row vector of size 1x3, also with its middle element highlighted in red. The result is a 4x3 matrix where every element in the first column is red, representing the outer product of the two vectors.

$$\begin{pmatrix} \color{red}\blacksquare \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \color{red}\blacksquare & \text{gray} \end{pmatrix} = \begin{pmatrix} \color{red}\blacksquare & \color{red}\blacksquare & \color{red}\blacksquare \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \end{pmatrix}$$

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product of two vectors. On the left, a column vector is shown with four elements: a red square at the top, followed by three gray squares. This is multiplied by a row vector with three elements: two gray squares followed by a red square. The result is a 4x3 matrix on the right, where each row is a scaled version of the row vector. The first row has a red square in the third column, while the other three rows have gray squares in all three columns.

$$\begin{pmatrix} \text{red} \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{gray} & \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} & \text{gray} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \end{pmatrix}$$

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T using square blocks to represent elements of matrices and vectors. On the left, a column vector a of size 4x1 is shown, with its second element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix where each row is a scaled version of the vector b^T . The first row of the resulting matrix has its first element highlighted in red, corresponding to the red elements in the input vectors.

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$$\begin{pmatrix} \square \\ \color{red}\square \\ \square \\ \square \end{pmatrix} \begin{pmatrix} \square & \color{red}\square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \color{red}\square & \square \\ \square & \square & \square \\ \square & \square & \square \end{pmatrix}$$

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a is shown with four elements: three gray squares and one red square in the second position. This is multiplied by a row vector b^T with three elements: two gray squares and one red square in the third position. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from a and b^T . The entire second row of the resulting matrix is red, representing the product of the red element in a and the row vector b^T .

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its third element highlighted in red. This is multiplied by a row vector of size 3, with its first element highlighted in red. The result is a 4x3 matrix where the third row of the first column is red, and all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \color{red}\square \\ \square \end{pmatrix} \begin{pmatrix} \color{red}\square & \square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \color{red}\square & \square & \square \\ \square & \square & \square \end{pmatrix}$$

Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product operation ab^T using square blocks to represent elements. On the left, a column vector a is shown with four gray blocks, where the third block from the top is red. This is multiplied by a row vector b^T shown in parentheses with three blocks: a gray block, a red block, and another gray block. An equals sign follows, leading to a 4x3 matrix on the right. This matrix is formed by scaling the row vector b^T by each element of the column vector a . The first and fourth rows consist of three gray blocks. The second row consists of three gray blocks. The third row, corresponding to the red block in a , consists of three red blocks.

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The diagram illustrates the outer product of two vectors. On the left, a column vector is shown with four elements: three gray squares and one red square in the third position. This is multiplied by a row vector with three elements: two gray squares and one red square in the third position. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from the two input vectors. The third row and third column of the resulting matrix are red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \color{red}\square \\ \square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \color{red}\square & \color{red}\square & \color{red}\square \\ \square & \square & \square \end{pmatrix}$$

Outer Products

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its bottom element highlighted in red. This is multiplied by a row vector of size 3, with its first element highlighted in red. The result is a 4x3 matrix where each row is a scaled version of the original row vector, and the first column is a scaled version of the original column vector. The red squares indicate the element-wise multiplication that forms the first row and first column of the resulting matrix.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \color{red}\square & \square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \color{red}\square & \square & \square \end{pmatrix}$$

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The diagram illustrates the outer product of two vectors. On the left, a column vector a of size 4x1 is shown, with its bottom element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its middle element highlighted in red. The result is a 4x3 matrix on the right, where the bottom row of the matrix has its middle element highlighted in red, corresponding to the product of the red elements in the input vectors. All other elements in the matrix are gray, representing the products of the other elements in the vectors.

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$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \square & \color{red}\square \end{pmatrix}$$

Outer Products

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The diagram illustrates the outer product of a column vector and a row vector. On the left, a column vector is shown with four elements: three gray squares and one red square at the bottom. This is multiplied by a row vector with three elements: two gray squares and one red square on the right. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from the input vectors. The bottom-right element is red, while all other elements are gray.

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- Notice that we need to know whether we are dealing with row or column vectors

Norms

- Norms are non-negative numbers associated with the “size” of a vector
- The norm you should be familiar with is the **Euclidean Norm** or 2-norm

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- We will see there exist other norms (using different ways of measuring size)
- The Euclidean norm is intimately related to the inner-product, we will see other examples of this for different vector spaces

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Matrices as Linear Mappings

- Matrices represent mappings between vectors

$$\mathbf{b} = \mathbf{M}\mathbf{a}$$

- The elements of $\mathbf{b}$ are **linear sums** of the elements of $\mathbf{a}$

$$b_i = \sum_j M_{ij} a_j$$

- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

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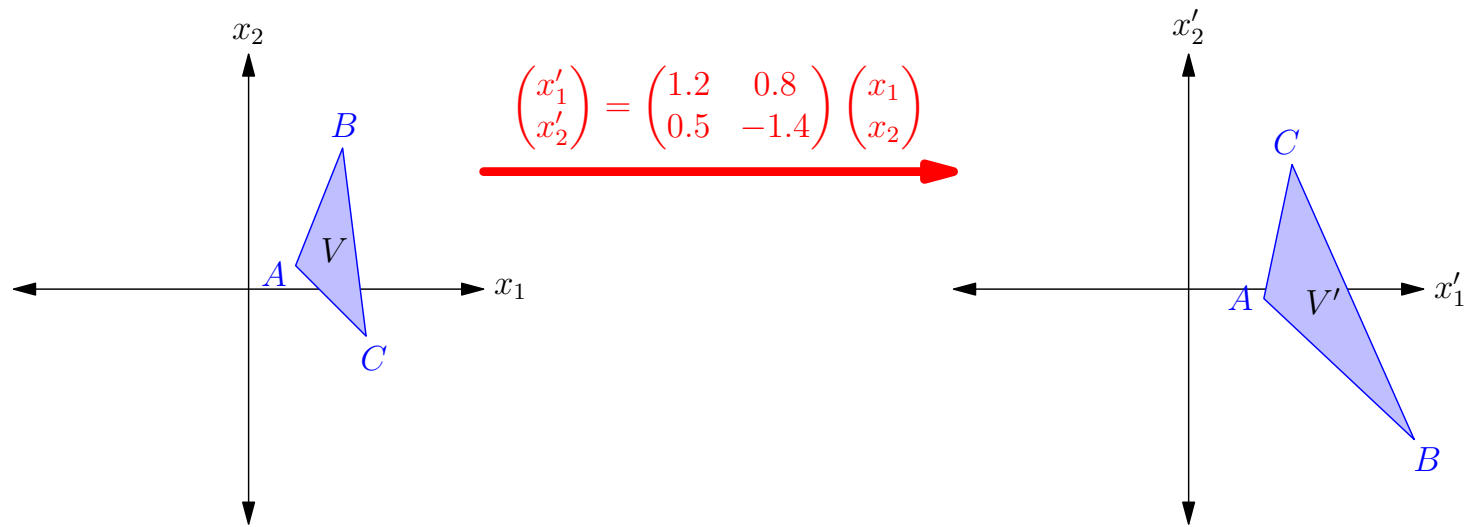
- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

$$\mathbf{M}(\mathbf{a} + \mathbf{b}) = (\mathbf{M}\mathbf{a}) + (\mathbf{M}\mathbf{b})$$

Action Of Matrices on Volumes

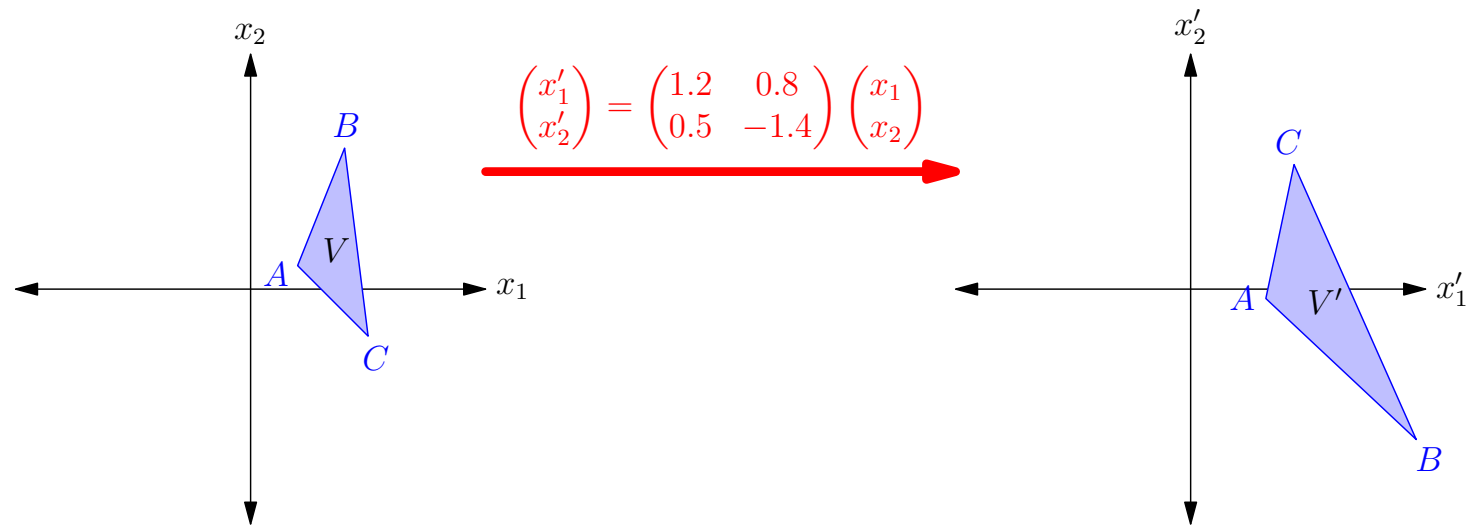
- For a closed set of points corresponding to a volume, V in $\mathbb{R}^n$, treating each point as a vector, we can ask what happens to the volume after mapping the vector by a matrix



- For square matrices, $\mathbf{M}$, the determinant is equal to $\det(\mathbf{M}) = \pm V'/V$
- The sign tells us if there has been a reflection

Action Of Matrices on Volumes

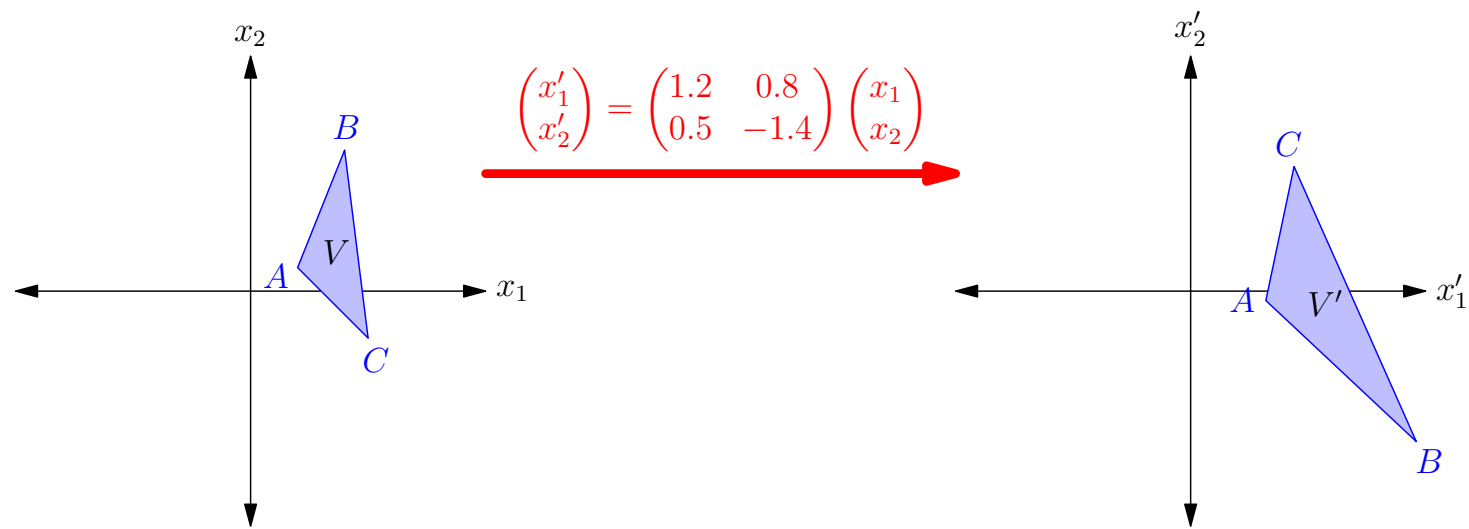
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Action Of Matrices on Volumes

- For a closed set of points corresponding to a volume, V in $\mathbb{R}^n$, treating each point as a vector, we can ask what happens to the volume after mapping the vector by a matrix



- For square matrices, $\mathbf{M}$, the determinant is equal to $\det(\mathbf{M}) = \pm V'/V$
- The sign tells us if there has been a reflection

Determinants

- A consequence of the fact that $\mathbf{AB}$ denotes the composition, apply mapping $\mathbf{B}$ then apply mapping $\mathbf{A}$

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \times \det(\mathbf{B})$$

- Determinants are only defined for square matrices, as non-square matrices project into different size spaces, where the volumes cannot be compared
- **Singular matrices** projects all points onto a lower dimensional space, so have determinant 0

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Multivariate Functions

- In machine learning we heavily use multivariate functions

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- Often our functions will be scalar value $f : \mathbb{R}^n \rightarrow \mathbb{R}$
- E.g. a learning machine for regression might be $\hat{f}(\mathbf{x}, \mathbf{w})$
- We say a function f is linear if

$$f(c\mathbf{x}) = cf(\mathbf{x}) \qquad f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$$

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Gradients

- The gradient of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at a point $\mathbf{x}^*$ is defined as the column vector

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

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- To reason about events with uncertain outcomes we introduce **random variables**
- Random variables corresponds to objects that we assign numbers to depending on the outcome of a random event
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- A continuous random variable might be the temperature at 12 noon tomorrow in Hyde park

Probability Distributions

- For discrete events, we can define a **probability mass function** $\mathbb{P}(X = x)$ that the random variable takes the value x
- This is a number between 0 and 1 with $\sum_{x \in \mathcal{X}} \mathbb{P}(X = x) = 1$
- For continuous events the probability of X taking a particular value x is zero
- Instead we consider the probability density $f_X(x)$ which we can define as

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x < X \leq x + \delta x)}{\delta x}$$

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Expectations

- One of the most useful quantities in dealing with random numbers is to look at the average value, aka as the **expectation**, for some function $g(X)$ of the random variable

$$\mathbb{E}[g(X)] = \begin{cases} \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) g(x) \\ \int f_X(x) g(x) dx \end{cases}$$

- Expectations are linear operators

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Lesson

- I've covered material that we assume you know
- If you don't, you are going to struggle following any machine learning course, so grab a maths book and revise this
- We will revisit most of these topics to gain a better understanding, but you will be lost in the other machine learning modules if you are not familiar with everything I've covered
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